The first derivative singlet in complex DSSYK
Two moments in the double-scaled limit

Alan Day Chengheng*

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Abstract

We evaluate the ensemble-centered equal-time strength and the second frequency moment of the first derivative singlet in complex DSSYK at infinite temperature. Fermionic Wick contractions reduce finite-N moments to coefficients of polynomials in at most six variables. Pairwise set overlaps then yield an explicit double-scaled result. The centered singlet has finite strength and finite initial evolution scale at fixed positive double-scaling parameter. Its normalized initial curvature tends to $16\mathcal{J}_s^2$ in the subsequent small-parameter limit.

1 A singlet with a finite dynamical scale

The Susskind programme motivates seeking singlet observables that can be matched to bulk propagation [1]. We use the same complex-SYK model as in our preceding notes:

$$H = \sum_{|I|=|J|=p} G_{IJ} c_I^\dagger c_J, \quad \mathbb{E}G_{IJ}G_{KL} = w\delta_{IL}\delta_{JK}, \quad w = 4^p \frac{\mathcal{J}_s^2}{\lambda}\left(\frac{p!}{N^p}\right)^2. \tag{1}$$

G is Hermitian, $\{c_i, c_j^\dagger\} = \delta_{ij}$, $\lambda = p^2/N$, $H = H_s$, and $\hbar = 1$. The state is the full $\beta = \mu = 0$ trace, $\mathrm{tr} = 2^{-N}\,\mathrm{Tr}$. Define

$$A = \sum_i [H, c_i]^\dagger [H, c_i], \qquad O = A - \mathbb{E}A, \qquad K = [H, A] = [H, O]. \tag{2}$$

Here $\mathbb{E}A = f(Q)$ is a deterministic function of particle number, not the expectation value in one realization. Let

$$V_0 = \mathbb{E}\,\mathrm{tr}\, O^2, \qquad M_2 = \mathbb{E}\,\mathrm{tr}\, K^\dagger K = \int \frac{\mathrm{d}\omega}{2\pi}\omega^2 S_O(\omega). \tag{3}$$

For $N, p \to \infty$ with $p^2/N = \lambda > 0$ fixed, our contraction calculation gives, with $q = e^{-4\lambda}$,

$$\boxed{V_0^{\mathrm{DS}} = 4\mathcal{J}_s^4\left(\lambda^{-1} + 1 + q\right),} \tag{4}$$

$$\boxed{M_2^{\mathrm{DS}} = \frac{8\mathcal{J}_s^6}{\lambda^2}\left[1 - q^2 + 4\lambda(q - q^2)\right].} \tag{5}$$

Both quantities are strictly positive for $\lambda > 0$. The large mean of A has been removed, while its dynamics is unchanged.

*AI research assistant (OpenAI).

2 What these moments tell us

For each finite system, the normalized centered correlator obeys

$$\frac{\mathbb{E}\,\mathrm{tr}\,O(t)O}{V_0} = 1 - \frac{1}{2}\Omega^2 t^2 + O_{N,p}(t^4), \qquad \Omega^2 = \frac{M_2}{V_0}, \tag{6}$$

the limiting initial curvature is

$$\Omega^2_{\mathrm{DS}} = \frac{2\mathcal{J}_s^2}{\lambda}\,\frac{1-q^2+4\lambda(q-q^2)}{1+\lambda(1+q)} \;\longrightarrow\; 16\mathcal{J}_s^2 \quad (\lambda \to 0^+). \tag{7}$$

Thus the candidate has a nontrivial initial evolution scale in string units. This supplies a quantitative microscopic input for the singlet-to-bulk programme. The next target is the frequency distribution that realizes these moments, followed by its comparison with the bulk Rindler propagator.

Exact finite-system information. The same calculation yields an exact coefficient formula for every N, p. In particular, for $p = 2$,

$$\frac{M_2(N,2)}{w^3} = \frac{N^2(N-3)(N-2)(N-1)(N+1)}{512} \times (N^4 + 4N^3 - 53N^2 + 120N - 16). \tag{8}$$

It reproduces 120 at $N = 4$ and 2700 at $N = 5$ in $w = 1$ units. An independent full covariance-tensor sum gives the latter value. For $p = 1$, the dynamic moment vanishes identically at every N.

Approach to the limit. The table compares exact finite-p coefficients with (4) and (5). These are ratios to the limiting expression, not fitted extrapolations. At $\lambda = 1/4$, $N = 4p^2$; at $\lambda = 1$, $N = p^2$.

p	V_0/V_0^{DS}, $\lambda = 1/4$	M_2/M_2^{DS}, $\lambda = 1/4$	M_2/M_2^{DS}, $\lambda = 1$
2	.940774	.691904	.218718
3	.931858	.803723	.535200
4	.937683	.851019	.595472
5	.944487	.878486	.628144

The deviations remain substantial for these small p. The limiting formulas follow from the overlap argument below, not from these data.

3 Route through the literature

We reuse fermionic Wick ordering [2] and the embedded-ensemble covariance structure [3, 4]. The known complex-SYK chord scale is $q = e^{-4\lambda}$ [5]. Particle-removal formulas in [4], Section 7, organize neighboring charge sectors, but retain coupled reduced matrix elements in the fourth mixed moment. For the present sixth-order coupling average, we instead perform an explicit mode-wise Wick reduction. This is an application of established contraction methods; no claim of mathematical or physical priority is made.

4 Exact reduction to six variables

Let $\mathcal{D}(X) = QX - \sum_i c_i^\dagger X c_i$. For a normal-ordered k-body monomial, $\mathcal{D}(X) = kX$. When normal ordering a product of two p-body monomials X, Y, multiply each contraction between the left annihilator block and the right creator block by $1 + h$. Call the resulting polynomial $R_h(X, Y)$. Then

$$R_0(X, Y) = XY, \qquad \partial_h R_h(H, H)|_0 = (2p - \mathcal{D})H^2 = A. \tag{9}$$

A term with r such contractions has body rank $2p - r$, proving (9) directly.

Use six ordered Hamiltonian slots. For two disjoint adjacent marked pairs $\mathcal{M} = ((r, s), (t, u))$, set

$$T_{ab}(h, g) = \begin{cases} \frac{1}{2}, & a \le b, \\ -\frac{1}{2}, & a > b, \end{cases} \quad \text{with } T_{sr} \mapsto T_{sr} - h, \quad T_{ut} \mapsto T_{ut} - g. \tag{10}$$

This is the signed thermal contraction matrix: a marked annihilator–creator contraction changes from $1/2$ to $1/2 + h$ or $1/2 + g$. For a coupling pairing π (a fixed-point-free involution), define

$$F_{\pi,\mathcal{M}}(z; h, g) = \sum_{S \subseteq \{1,\dots,6\}} \det[T_{a,\pi(b)}]_{a,b \in S} \prod_{a \in S} z_a = \det(I + ZTP_\pi), \tag{11}$$

where $Z = \mathrm{diag}(z_1, \dots, z_6)$ and $(TP_\pi)_{ab} = T_{a,\pi(b)}$. The empty determinant is one. Let $[z^p]$ mean the coefficient of $z_1^p \cdots z_6^p$. The contribution of a fixed paired word is

$$w^3 (-1)^{3p} [z^p] F_{\pi,\mathcal{M}}(z; h, g)^N. \tag{12}$$

All overlaps $I \cap J$, including repeated mode labels in a vertex, are included. No Fock-space cutoff or random-sample average is used.

Write $K^\dagger K = AHHA - AHAH - HAHA + HAAH$ before interpreting each A as a marked pair. The four choices are

	$\mathcal{M}$	$\epsilon_\mathcal{M}$
$AHHA$	$((1,2),(5,6))$	$+1$
$AHAH$	$((1,2),(4,5))$	-1
$HAHA$	$((2,3),(5,6))$	-1
$HAAH$	$((2,3),(4,5))$	$+1$

(13)

Only pairings connecting the first three slots to the last three are needed: the preceding note proves that K has no linear Gaussian component. Denote these six pairings by $\mathcal{P}_\times$.

$$\boxed{\frac{M_2}{w^3} = (-1)^{3p} [z^p h g] \sum_{\pi \in \mathcal{P}_\times} \sum_{\mathcal{M}} \epsilon_\mathcal{M} F_{\pi,\mathcal{M}}^N.} \tag{14}$$

A Signs, finite coefficients and independent checks

For a pairing π, choose six independent creator sets I_a, each of size p; the annihilator set at slot b is $I_{\pi(b)}$. Ordinary thermal Wick contraction is a determinant with $6p$ rows and columns. Permute the annihilator blocks to the order matching the creator blocks. The permutation has sign $(-1)^{p^2(6/2)} = (-1)^{3p}$. Grouping both rows and columns by mode then gives identical further permutation signs, whose product is one. The determinant factors into the local minors in (11). Summing the possible membership subsets of each of the N modes gives F^N, and $[z^p]$ enforces every set size. This proves (12).

For integer computation replace T by $2T$, obtaining G with integer coefficients. The denominator of the extracted term is always 2^{6p}. Since $G-1$ has no constant term,

$$\frac{M_2(N,p)}{w^3} = \frac{1}{2^{6p}} \sum_{j=0}^{6p} \binom{N}{j} c_j(p), \tag{15}$$

where the signed c_j are coefficients of $(G-1)^j$, summed over the 24 marked configurations. For a fixed p, these coefficients are computed once and then evaluated at any N using exact integers.

An independent formula keeps all 15 Gaussian pairings in $2\,\mathrm{tr}(H^2A^2 - HAHA)$. Its coefficient arrays agree exactly with (15) for $p = 1, 2, 3$. The $p = 1$ array is zero. Arrays through $p = 5$ vanish for all $N \le p+1$, as required by the conserved controls. For $N = 5, p = 2$, direct Gaussian tensor contraction gives 2700, agreeing with (8).

The local marked-word formula itself was checked before ensemble summation: 720 selected words at $(N,p) = (4,1), (5,2), (6,3)$ compare all four coefficients $1, h, g, hg$ with direct CAR matrices. All 2880 rational comparisons agree, including nonzero negative words. The selected index sets and seed are retained; these are algebra tests, not disorder samples used to estimate a moment.

B The double-scaled overlap limit

For a cross pairing, the singleton coefficients of F are $a_i = +1/2$ for $i = 1, 2, 3$ and $a_i = -1/2$ for $i = 4, 5, 6$. None is affected by our marks. Put

$$W_S = \frac{\det[T_{a,\pi(b)}]_{a,b\in S}}{\prod_{a\in S} a_a}, \qquad W_\varnothing = W_{\{i\}} = 1. \tag{16}$$

The coefficient in (12), divided by $\binom{N}{p}^6 \prod_i a_i^p$, is exactly $\mathbb{E}_{I_1,\dots,I_6} \prod_{\ell=1}^N W_{S_\ell}$ for six independent uniform p-subsets of $\{1,\dots,N\}$.

Triple membership has probability at most $\binom{6}{3}p^3/N^2 = o(1)$. The 15 pairwise overlap counts converge jointly to independent Poisson variables of mean λ. One direct proof uses joint factorial moments: assignments with distinct sites give $(N)_r \prod_i (p)_{d_i}/(N)_{d_i} \to \lambda^r$, where $\sum_i d_i = 2r$; shared sites have higher membership and vanish. This argument concerns a fixed number of sets, not a number growing with N.

For completeness, convergence also holds for the weighted expectation, not just in distribution. On a bounded complex neighborhood of $h = g = 0$, the finitely many W_S obey $|W_S| \le R^{\binom{|S|}{2}}$ for some $R \ge 1$. If $Y = \sum_{i<j} |I_i \cap I_j|$, then $|\prod W_S| \le R^Y$. Hypergeometric factorial moments and Hölder's inequality give a uniform bound on $\mathbb{E}e^{cY}$ for any fixed c when p^2/N remains in a compact interval. For one overlap X, for instance,

$$\mathbb{E}e^{cX} \le \exp\left[\frac{p^2}{N-p+1}(e^c - 1)\right], \qquad c \ge 0. \tag{17}$$

Cauchy–Schwarz bounds the contribution of triple membership by a constant times $\sqrt{\binom{6}{3}p^3/N^2}$, which tends to zero. The same exponential-moment bound gives uniform integrability of the pair weights. Consequently the convergence is locally uniform in complex h, g, and Cauchy's formula permits the two marked derivatives to pass to the limit.

Write b_{ij} for the coefficient of $z_i z_j$ in F. We obtain

$$F^N\big|_{[z^p]} \sim \frac{N^{6p}}{(p!)^6} \prod_i a_i^p \, \exp\left\{\lambda\left[-3 + \sum_{i<j}\left(\frac{b_{ij}}{a_i a_j} - 1\right)\right]\right\}, \tag{18}$$

where this denotes convergence of the ratio by the nonzero displayed prefactor, uniformly in the marking variables. With zero marks the exponent is $-4\lambda\,\mathrm{cr}(\pi)$, reproducing the known complex-SYK chord weight [5]. In general it is

$$-4\lambda\,\mathrm{cr}(\pi) + \lambda(\alpha h + \beta g + \delta hg). \tag{19}$$

After the two derivatives, each graph contributes

$$\frac{\mathcal{J}_s^6}{\lambda^3} q^{\mathrm{cr}(\pi)}(\lambda\delta + \lambda^2\alpha\beta). \tag{20}$$

In this normalization the factors $(-1)^{3p}$ and $\prod_i a_i^p$ cancel their signs, and w^3 cancels the factorial and N powers.

C The small graph sum

The following table gives the coefficients after summing the four commutator words, for each pairing. A tuple (a, b, c) means that slots $1, 2, 3$ are paired respectively with a, b, c.

Pairing	crossings	$\sum \epsilon\delta$	$\sum \epsilon\alpha\beta$
$(4, 5, 6)$	3	0	0
$(4, 6, 5)$	2	-4	-16
$(5, 4, 6)$	2	-4	-16
$(5, 6, 4)$	1	0	16
$(6, 4, 5)$	1	0	16
$(6, 5, 4)$	0	8	0

Their sum is

$$\frac{\lambda^3 M_2^{\mathrm{DS}}}{\mathcal{J}_s^6} = 8\lambda(1 - q^2) + 32\lambda^2(q - q^2), \tag{21}$$

which proves (5).

The centered equal-time strength requires only four slots, with marks $((1, 2), (3, 4))$. Subtracting $\mathbb{E}A$ removes the Gaussian pairing within each marked pair. Of the two remaining pairings, the crossing one has $(\alpha, \beta, \delta) = (-2, -2, 0)$; the nested one has $(2, 2, 4)$. The identical four-set overlap argument gives

$$\frac{\lambda^2 V_0^{\mathrm{DS}}}{\mathcal{J}_s^4} = 4\lambda^2 q + 4\lambda + 4\lambda^2, \tag{22}$$

proving (4). Its finite coefficient calculation reproduces $V_0(N, 1)/w^2 = N(N + 1)^2/4$ and $V_0(4, 2)/w^2 = 483/4$.

D Interpretation and limits

These are model calculations for the full complex Gaussian ensemble, with standard CAR and all overlapping index sets retained. The order of limits in (7) is double scaling at fixed positive λ, followed by $\lambda \to 0^+$. Keeping p fixed instead gives a different limit; the exactly conserved $p = 1$ model is a simple example. No uniform error estimate for a simultaneous $\lambda(N) \to 0$ limit is asserted.

The Gaussian-centered O can still overlap with sample-dependent conserved operators. Thus Ω is the initial curvature scale of this specified correlator, not an isolated particle mass, a late-time decay rate, or a gap. The two moments do not determine the full spectral density or establish convergence of the entire time correlator. Equation (7) is a limit of finite-system curvatures; interchange of the time and double-scaling limits requires further control. In cosmic units $H_c = pH_s$ and $A_c = p^2 A_s$, $V_{0,c} = p^4 V_{0,s}$ and $M_{2,c} = p^6 M_{2,s}$.

E Reproducibility and next calculation

The integer coefficient engine and its drivers are in the code directory:

```
singlet_coefficient.py
check_singlet_coefficient.py
run_singlet_double_scale.py
run_singlet_centered_moment.py
```

The stage results retain exact arrays through $p = 5$, factored polynomials, all marked words, a separate full-tensor check, and the 24-graph table. The numerical environment is Python 3.14.5, NumPy 2.4.6, SymPy 1.14.0. Integer coefficient extraction has no statistical error or spectral tail cutoff. The displayed finite-p ratios are rounded only for reading.

An initial overlap audit omitted the -1 contribution of identically zero minors when computing $\sum_{i<j}(b_{ij}/a_i a_j - 1)$. Its assertion failed and the original implementation was retained. The repaired audit includes all 15 pairs and checks their known chord exponents. The finite coefficient extraction and marked derivatives were unaffected. The separate finishing script records the corrected checks without restarting completed coefficient calculations.

The next calculation is a controlled construction of further moments or the energy-resolved singlet kernel. The finite polynomial method extends to longer words, while the marked overlap rules provide the double-scaled simplification. We will first determine which part of V_0 remains at zero frequency and compute the fourth frequency moment; these are the next inputs needed to constrain the propagating spectrum.

References

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